Tool lifecycle

How a screener tool is added, started, paused, stopped, deleted and restored, and what happens underneath each button of **Manage tools** (`/screeners/manage`). Written from `src/toolsManager.js`, `src/toolsRegistry.js`, `src/routes/toolsManage.js`, `deploy/tool-ctl.sh`, `src/archive/server.js` and `src/sideA/toolIdentity.js`.

Related: how to use the page (`docs/USER-GUIDE.md`), the whole system (`docs/ARCHITECTURE.md`), the routes (`docs/API.md`).

What a tool is

Every tool is **its own program**: its own pm2 process, port, database, scorer and memory ceiling, so a broken tool cannot take another down with it.

	T1	**any other tool T*n***
pm2 process	`tool-T1`	`tool-Tn`
scorer process (when it has one)	`scorer-T1` on port + 1	`scorer-Tn` on port + 1
database	`data/tradedesk.db`	`data/tn.db`
model folder	`src/scoring/outputs`	`src/scoring/outputs-tn`
temp folder	`tmp`	`tmp-tn`

The process gets `TOOL_ID`, `TOOL_NAME`, `PORT`, `DB_PATH`, `MODEL_OUTPUT_ROOT`, `TMP_DIR` and `SCORER_URL` from `deploy/tool-ctl.sh`.

T1 is special. It serves the landing page and Manage tools, so it can be neither stopped nor deleted, and the Manage tools routes exist on T1 only.

The registry: which tools exist

Two files, merged by `src/toolsRegistry.js`:

File	Holds	Written by
`tools.config.json` (in the repo)	the built-in tools: id, name, port, flags, description	a code change
`data/tools.local.json` (on the box)	tools added from the page, and overrides of any tool	Manage tools

Everything that lists tools (the landing page, the archive, qp, the deploy) reads the merged result, so they cannot disagree.

Overrides may change only `name`, `desc`, `accent`, `captureAt` and `enabled`, plus the markers `deleted` and `pendingRestart`. Ports and ids belong to the registry. The file is written to a temp file and renamed, so a crash never leaves half a file.

Three flags, three questions:

Flag	Means	Absent means
`enabled`	the deploy starts it; it scans	true
`archive`	when stopped, the archive serves its stored registers on its port	false for built-in tools; true for tools added from the page
`scorer`	it has a Python scoring process	true (added tools: false)

Merge rules: a deleted tool is always disabled and archived; a tool switched on is never archived (it holds its own port); a tool switched off is archived if it was added from the page or its override does not say otherwise.

Adding a tool

`POST /api/tools/manage` with `name`, and optionally `desc`, `accent`, `captureAt`.

1. **Limit:** at most **6** added tools alive (deleted ones do not count).
2. **Id:** one past the highest id ever used, deleted tools included, so an id and the history filed under it are never reused.
3. **Port:** the next free multiple of ten past every port in use by a tool, its scorer or an app; the scorer port is port + 1.
4. **Defaults:** no scorer; the next accent colour in the palette; capture times 09:36 / 09:37 / 09:40.
5. **Start:** if there is enough memory (below), it is started at once; otherwise it is created stopped and the message says why.
6. qp is told to re-read its list of sources.

The new tool has **no screeners**; you build them on its own page.

Capture times (`captureAt`)

`{ r1, entryA, entryB }`, each `HH:MM`:

- they must run in order: `r1 < entryA < entryB`;
- inside the session: `r1 ≥ 09:30`, `entryB ≤ 15:50`;
- an optional `why` note is kept with them.

The buttons

Button	Route	What happens
Start	`POST /:id/start`	refused if deleted (restore first) or if memory is short. Marks it enabled; if the archive was serving it, the archive is restarted first so it lets go of the port; runs `tool-ctl.sh start`. If that fails, it is marked disabled again and the archive restored. Then qp re-reads its sources
Stop	`POST /:id/stop`	not T1. `tool-ctl.sh stop` (deletes the pm2 process and its scorer), marked disabled, any pending restart cleared, the archive restarted so it now serves this tool's history, qp re-reads
Pause / **Resume**	`POST /:id/pause` / `resume`	calls the tool's own `/api/tool/pause` or `resume`. The program stays up; it stops **scanning** only (below)
Scan now	`POST /:id/scan`	calls the tool's `/api/scan/run`. Waits 4 s; if the scan is still going it says "is scanning", otherwise how many rows, or that the tool is paused
Edit	`PUT /:id`	name, description, colour, capture times. A new name is sent to the running tool at once; a new name or colour makes qp re-read. Capture times: see the session rule
Delete	`DELETE /:id`	not T1. Stops it, marks it deleted and disabled (off every list), restarts the archive so its history stays readable, qp re-reads. **Nothing is erased**
Restore	`POST /:id/restore`	takes the deleted mark off; it comes back **stopped** — press Start

All routes are under `/api/tools/manage` on T1. Each answers `{ ok, message, … }`, or `400 { ok: false, error }` with the reason.

The two guards

1. **Memory.** A tool is not started (by Start or Add) with less than **180 MB** available (`TOOLS_MIN_FREE_MB`), read from `/proc/meminfo`. On a 912 MB box the alternative is the kernel choosing what to kill, and it may choose sshd. If memory cannot be read, the start goes ahead.
2. **The session rule.** Capture times are read when a tool starts, so a change needs a restart, and a restart empties the list the tool has built since 04:00. So a capture-time change on a running tool:
   - **04:00–16:30 ET on a weekday:** saved, marked `pendingRestart`, and applied at **16:50** by T1's job "Apply tool schedule changes", after the last capture (16:05) and training (16:20);
   - **any other time:** the tool is restarted at once.

   T1 restarting itself runs the command detached, so it survives its own restart.

Pause in detail

Stored in the tool's own database (`tool.paused` = `{ at, reason }`), so a pause survives a restart.

- A paused tool's discovery scans and refreshes return at once, saying it is paused (`docs/MONITOR.md`).
- Cards already found stay on screen; the registers are not touched.
- Its status (`/api/scan/status`, `/api/monitor`, `/api/tool`) says paused, since when and why, so a quiet tool is never mistaken for a broken one.

Why pause exists: the tools are separate experiments; pausing one stops it from collecting more without destroying what it has.

`deploy/tool-ctl.sh`

The one place a tool is started. `deploy-tools.sh` calls it for every enabled tool, and Manage tools calls it for the one pressed, so a tool started from the phone gets exactly what a deploy gives it.

```
bash deploy/tool-ctl.sh start   T13
bash deploy/tool-ctl.sh stop    T13
bash deploy/tool-ctl.sh restart T13
bash deploy/tool-ctl.sh archive
```

Action	Does
`start` / `restart`	looks the tool up in the merged registry; stops any old process; starts `scorer-Tn` (unless `scorer: false`) and `tool-Tn` with the tool's environment and memory cap; `pm2 save`
`stop`	deletes `tool-Tn` and `scorer-Tn`; `pm2 save`
`archive`	restarts the single `archive` process to serve every tool that is archived and disabled; with none, it does not run

Details that matter:

- **A clean environment.** Run from inside a tool, the script would inherit that tool's pm2 settings (its name, id, log paths, `TOOL_ID`, `PORT` …). It clears them first, so a tool started from T1's page gets nothing of T1's.
- **Memory caps** (pm2 restarts the process above them):

Process	Cap
T1	280M
T2, T6	240M
T11	200M
T7	180M
any other tool	140M (`TOOL_MAX_MEM`)
a scorer	180M (`SCORER_MAX_MEM`)
the archive	120M (`ARCHIVE_MAX_MEM`)

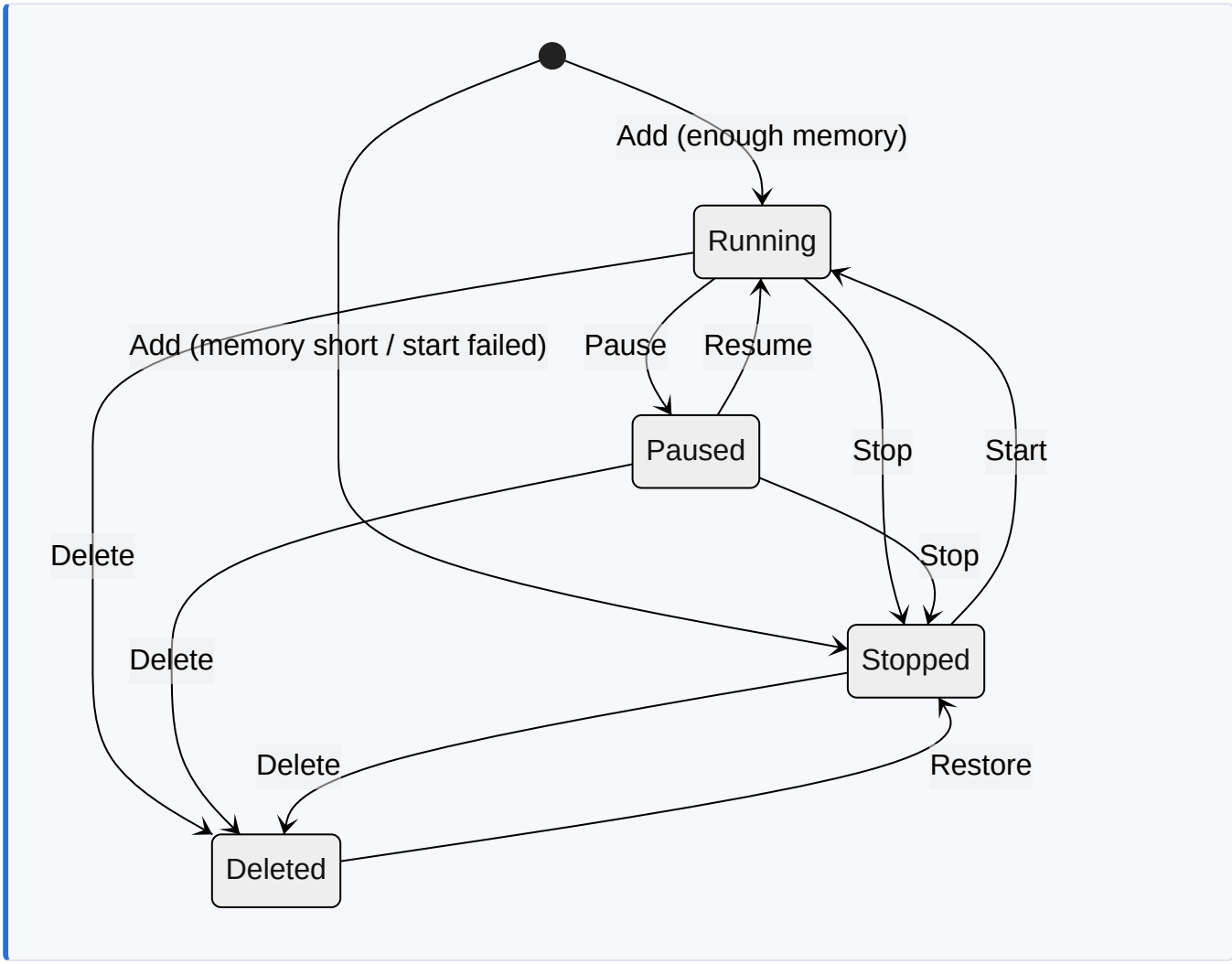
  A single tool's cap can be set with `TOOL_MAX_MEM_<ID>`.
- `TOOL_CTL_NO_SAVE` skips `pm2 save` (the deploy saves once at the end).
- It does not pull, install or touch any other process.

The archive

One read-only process (`src/archive/server.js`, about 45 MB) opens each archived tool's database and answers the same `/api/warehouse/*` routes on **that tool's own port**. qp reads tools over HTTP, so from qp's side a stopped tool is unchanged: same URL, same JSON. Without it, stopping a tool would take every chart and backtest of its history with it.

It has no scanner, scheduler, scorer or write route. It is restarted (`tool-ctl.sh archive`) whenever the set of stopped tools changes: on Stop, Delete, and before Start of an archived tool, so it releases that port.

The life of a tool, end to end



State	Program	Scans	History readable
Running	up	yes	yes, by the tool
Paused	up	no	yes, by the tool
Stopped	down	no	yes, by the archive (if archived)
Deleted	down, hidden from lists	no	yes, by the archive